

COMPREHENSIVE LIFE EXPECTANCY PREDICTION SYSTEM USING MULTIPLE LINEAR REGRESSION

*A project report
submitted in partial fulfilment of the requirements*

for

**SEP 786 – ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING FUNDAMENTALS**

TEAM MEMBERS	STUDENT ID
HARIKRISHNA RAMANATHAN BALAKRISHNAN	400547177
LAVANYA SINGH	400547269
SURAJ RAMESH	400547169
VIPIN CHANDRAN MUTHIRIKKAPARAMBIL	400547434

ENGINEERING
W Booth School of Engineering
Practice and Technology



14 NOVEMBER, 2023

INDEX

ABSTRACT

LIST OF FIGURES

ACKNOWLEDGEMENT

1	INTRODUCTION	1
1.1	MOTIVATION	1
2	METHODOLOGY	
2.1	DATA COLLECTION	2
2.1.1	VARIABLE DESCRIPTIONS	2-3
2.2	DATA PREPROCESSING	
2.2.1	DEALING WITH MISSING VALUES	3
2.2.2	DEALING WITH OUTLIERS	5
2.2.3	DEALING WITH CATEGORICAL FEATURES	7
2.3	EXPLORATORY DATA ANALYSIS	8-12
3	MODEL PREDICTION / REGULARIZATION AND EVALUATION METRICS	
3.1	TRAIN – TEST – SPLIT	12
3.2	MODEL FIT	12
3.3	MEAN ABSOLUTE ERROR	12
3.4	MEAN SQUARED ERROR	13
3.5	ROOT MEAN SQUARED ERROR	13
3.6	R ² SCORE	13
3.7	RIDGE REGRESSION	13
3.8	LASSO REGRESSION	14
3.9	ELASTIC NET	14
4	DEPLOYMENT AS WEB APP USING STREAMLIT	
4.1	VIRTUAL ENVIRONEMENT ANACONDA	15
4.2	SAVING TRAINED MODEL AS .CSV FILE	15
4.3	USING SPYDER IDE TO BUILD USER INTERFACE	15

ABSTRACT

Life expectancy stands as a pivotal statistical metric, representing the average duration a human being is anticipated to live. This project focuses on understanding and analysing the multifaceted factors that significantly impact global life expectancy rate. The proposed solution is to build a '**Multiple Linear Regression model**' to predict the "Global General Life Expectancy" based on predicting variables divided into several broad categories: Immunization related factors, Mortality factors, Economical factors and Social factors. Since the observations in this dataset are based on different countries, it will be easier for a country to determine the predicting factor which is contributing to lower value of life expectancy. This will also help in suggesting a country in which evidence-based public health policies, precision medical interventions, and targeted social programs should be given importance in order to efficiently improve the life expectancy of its population.

LIST OF FIGURES

Fig No	Description
1	Reading values from CSV file and converting them to a dataframe.
2	Finding the null values in each feature.
3	Finding out the outliers in the given dataset using Boxplot
4	Correlation Matrix Heatmap
5	Scatterplot of Schooling vs Life Expectancy
6	Scatterplot of Income Composition vs Life Expectancy
7	Finding out the outliers in the given dataset using Boxplot
8	Applying IQR (Inter Quartile Range) on the dataset to remove any outliers
9	Different categories of data as seen in Status feature: Developed and Developing
10	One Hot Encoding (OHE) performed on countries and status
11	Developing countries status shown as 1 post OHE
12	Histogram showing the distribution curve for life expectancy
13	Plot showing the relation between Life Expectancy and Status of the country
14	Plot showing the relation between Adult Mortality and Life Expectancy
15	Plot showing the relation between Infant deaths and Life Expectancy
16	Plot showing the relation between Schooling and Life Expectancy
17	Plot showing the relation between Life Expectancy and Immunization Coverage (Polio)
18	Plot showing the relation between Life Expectancy and Alcohol Consumption.
19	Plot showing the relation between Life Expectancy and Total Expenditure
20	Heatmap showing the relation between Life Expectancy, Schooling, BMI, Alcohol and Expenditure
21	Fitting the Linear Regression Model onto the Training Data
22	Evaluation Metrics performed
23	Ridge Regression
24	Lasso Regression
25	Elastic Net
26	StreamLit loaded into Anaconda Environment
27	Saving the trained model as a .csv file
28	User Interface of the Web App deployed using StreamLit

ACKNOWLEDGEMENT

We take this opportunity to express our deepest sense of gratitude and sincere thanks to everyone who helped us to complete this work successfully. We express our sincere thanks to Sayyed Faridoddin Afzali, our course instructor (SEP 786 Artificial Intelligence and Machine Learning Fundamentals) for providing us with all the necessary facilities and support. The data was collected from WHO and United Nations website with the help of Deeksha Russell and Duan Wang.

Harikrishna Ramanathan Balakrishnan
Lavanya Singh
Suraj Ramesh
Vipin Chandran Muthirikkaparambil

1. INTRODUCTION

One of the most important measurement methods is life expectancy, which is the average number of years that people are expected to live. The goal of this study is to comprehend and analyse the complex factors that have a major impact on the rate of life expectancy worldwide. Through the application of this concept, nations can pinpoint particular elements that lead to reduced life expectancy among their citizens.

This realisation is crucial for developing evidence-based public health policies, tailored social programmes, and precise medical therapies. The objective is to equip nations with the necessary knowledge to strategically concentrate on areas that will most effectively raise the average life expectancy of their people.

1.1 MOTIVATION

The data-set analysis, predictions aims to answer the following key questions:

- ☐ Do various predicting factors which have been chosen initially really affect the Life expectancy? What are the predicting variables actually affecting life expectancy?
- ☐ Should a country having a lower life expectancy value (<65) increase its healthcare expenditure in order to improve its average lifespan?
- ☐ How do Infant and Adult mortality rates affect life expectancy?
- ☐ Does Life Expectancy have positive or negative correlation with eating habits, lifestyle, exercise, smoking, drinking alcohol etc.
- ☐ What is the impact of schooling on the lifespan of humans?
- ☐ Does Life Expectancy have a positive or negative relationship with drinking alcohol?
- ☐ Do densely populated countries tend to have lower life expectancy?
- ☐ What is the impact of Immunization coverage on life Expectancy

This project, fuelled by these compelling questions, offers the prospect of uncovering insights that can inform policy, shape public health initiatives, and ultimately contribute to the collective goal of enhancing human life expectancy worldwide. It is a journey driven by the innate curiosity to decipher the intricacies of our existence and empower societies to make informed decisions for a healthier, more resilient future.

2. METHODOLOGY

2.1 DATA COLLECTION

<https://www.kaggle.com>

The project relies on accuracy of data. The Global Health Observatory (GHO) data repository under World Health Organization (WHO) keeps track of the health status as well as many other related factors for all countries. The data-sets are made available to the public for the purpose of health data analysis. The data-set related to life expectancy, health factors for 193 countries has been collected from the same WHO data repository website and its corresponding economic data was collected from the United Nation website. Among all categories of health-related factors only those critical factors were chosen which are more representative. It has been observed that in the past 15 years, there has been a huge development in the health sector resulting in improvement of human mortality rates especially in the developing nations in comparison to the past 30 years. Therefore, in this project we have considered data from the years 2000-2015 for 193 countries for further analysis. As the data-sets were from WHO, we found no evident errors. All predicting variables were then divided into several broad categories: Immunization related factors, Mortality factors, Economical factors and Social factors.

2.1.1 VARIABLE DESCRIPTIONS

- ❑ **country** (Nominal) - the country in which the indicators are from (i.e. United States of America or Congo)
- ❑ **year** (Ordinal) - the calendar year the indicators are from (ranging from 2000 to 2015)
- ❑ **status** (Nominal) - whether a country is considered to be 'Developing' or 'Developed' by WHO standards
- ❑ **life_expectancy** (Ratio) - the life expectancy of people in years for a particular country and year
- ❑ **adult_mortality** (Ratio) - the adult mortality rate per 1000 population (i.e. number of people dying between 15 and 60 years per 1000 population); if the rate is 263 then that means 263 people will die out of 1000 between the ages of 15 and 60; another way to think of this is that the chance an individual will die between 15 and 60 is 26.3%
- ❑ **infant_deaths** (Ratio) - number of infant deaths per 1000 population; similar to above, but for infants
- ❑ **alcohol** (Ratio) - a country's alcohol consumption rate measured as litres of pure alcohol consumption per capita
- ❑ **percentage_expenditure** (Ratio) - expenditure on health as a percentage of Gross Domestic Product (gdp)
- ❑ **hepatitis_b** (Ratio) - number of 1 year olds with Hepatitis B immunization over all 1 year olds in population
- ❑ **measles** (Ratio) - number of reported Measles cases per 1000 population
- ❑ **bmi** (Interval/Ordinal) - average Body Mass Index (BMI) of a country's total population
- ❑ **under-five_deaths** (Ratio) - number of people under the age of five deaths per 1000 population

- ❑ **polio** (Ratio) - number of 1 year olds with Polio immunization over the number of all 1 year olds in population
- ❑ **total_expenditure** (Ratio) - government expenditure on health as a percentage of total government expenditure
- ❑ **diphtheria** (Ratio) - Diphtheria tetanus toxoid and pertussis (DTP3) immunization rate of 1 year olds
- ❑ **hiv/aids** (Ratio) - deaths per 1000 live births caused by HIV/AIDS for people under 5; number of people under 5 who die due to HIV/AIDS per 1000 births
- ❑ **gdp** (Ratio) - Gross Domestic Product per capita
- ❑ **population** (Ratio) - population of a country
- ❑ **thinness_1-19_years** (Ratio) - rate of thinness among people aged 10-19 (Note: variable should be renamed to thinness_10-19_years to more accurately represent the variable)
- ❑ **thinness_5-9_years** (Ratio) - rate of thinness among people aged 5-9
- ❑ **income_composition_of_resources** (Ratio) - Human Development Index in terms of income composition of resources (index ranging from 0 to 1)
- ❑ **schooling** (Ratio) - average number of years of schooling of a population

2938 rows x 22 columns

Fig 1. Reading values from CSV file and converting them to a dataframe.

2.2 DATA PRE-PROCESSING

2.2.1 DEALING WITH MISSING VALUES

It has 22 columns with 20 Quantitative and 2 Qualitative features and nearly 3000 entries and with the inequality in no. of observations from each feature it is clear that it has got some null values.


```

In [18]: df_copy.isna().sum()
Out[18]: Country          0
Year          0
Status         0
Life expectancy  0
Adult Mortality  0
infant deaths   0
Alcohol        194
percentage expenditure  0
Hepatitis B     553
Measles         0
BMI             34
under-five deaths  0
Polio          19
Total expenditure 226
Diphtheria      19
HIV/AIDS        0
GDP            448
Population      652
thinness 1-19 years 34
thinness 5-9 years 34
Income composition of resources 167
Schooling       163
dtype: int64

```

Fig 2. Finding the null values in each feature.

There are many columns with null values but the amount of missing values is not big enough to drop the columns. So, imputing the missing values would be a good idea. We also know that all the columns with missing values are numerical continuous variables.

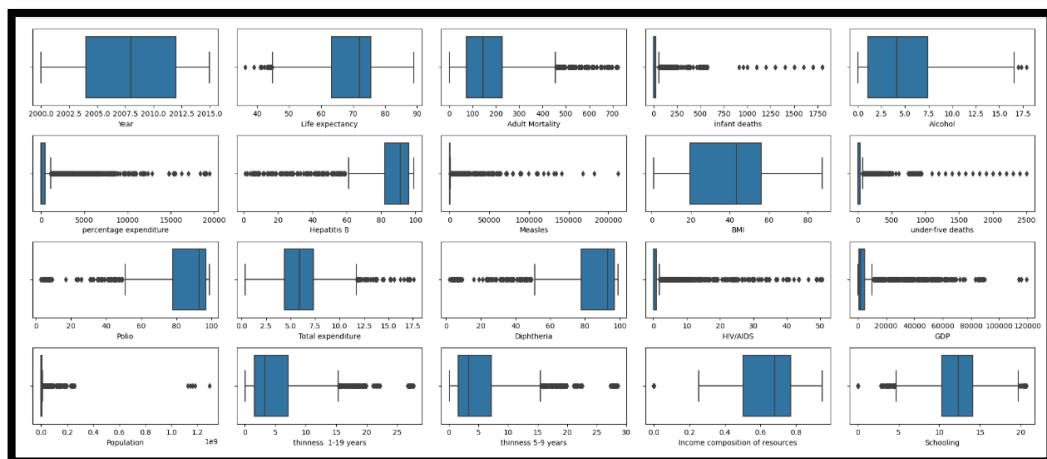


Fig 3. Finding out the outliers in the given dataset using Boxplot

Filling missing values with central tendency mean would not be a good idea because of the outliers. We can also fill it with the median but since data is divided into countries by years the median value will vary country wise. The appropriate solution would be to interpolate the values using the country.

It turned out that interpolation didn't fill the missing values. There are two reasons behind it.

1. Countries data for all the null values are null for each year.
2. Many countries have the first value as null and this method doesn't fill the first null entry.

So, the next possible method would be to fill the missing values using **Median but Year wise**.

2.2.2 DEALING WITH OUTLIERS

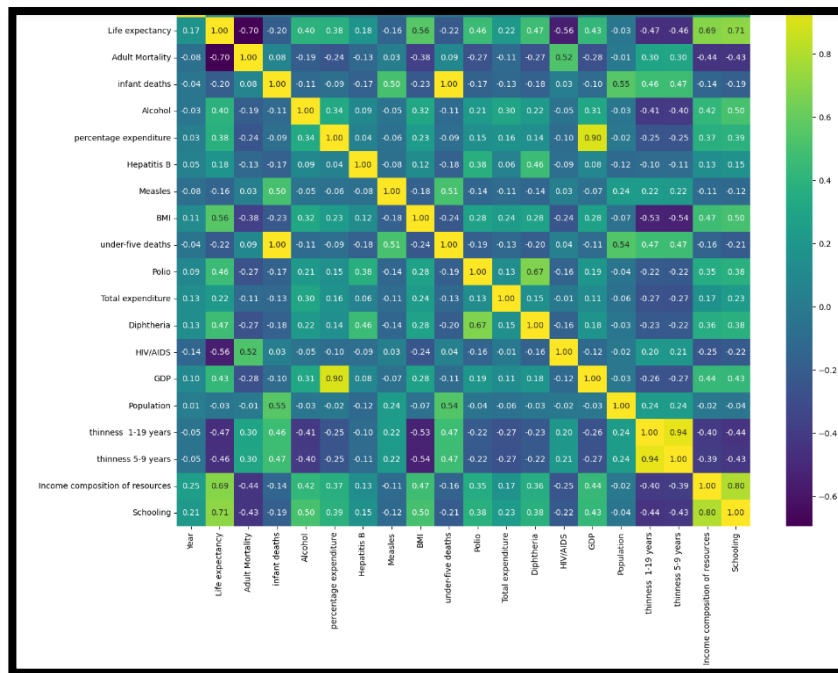


Fig 4. Correlation Matrix Heatmap

From the correlation matrix heatmap it looks like "Under-five death" and "infant deaths" are the same columns since their correlation is = 1. so we can remove one of them to remove redundant information introduced into the model being built.

Based on the correlation matrix heatmap let's analyse the nature of outliers. Based on the correlation values, on comparing highly correlated features. For, Schooling < 0.1 & Life expectancy > 0, there are 26 data points of outliers, in this comparison but here since the data is specific country wise and year wise we cannot consider this as outliers, because there may be cases where people without schooling i.e. our older generations have had high life expectancies

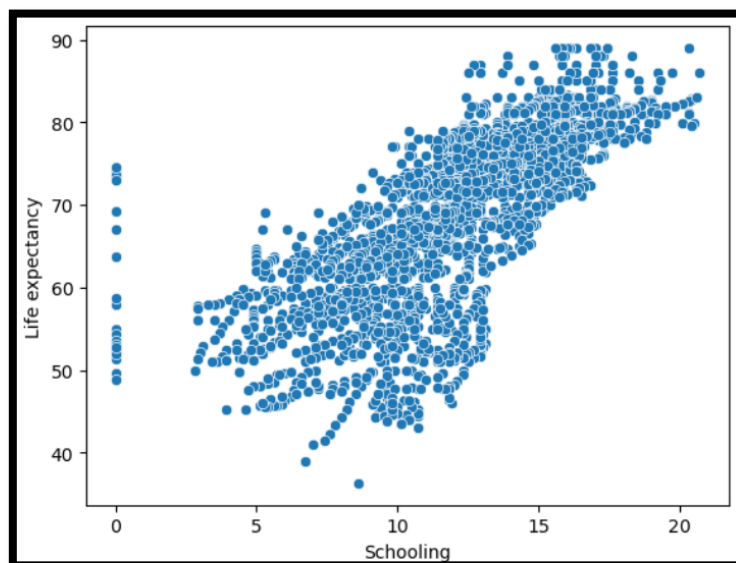


Fig 5. Scatterplot of Schooling vs Life Expectancy

Similarly for Income composition of resources < 0.1 & Life expectancy > 0 , we can notice similar situation, but still we cannot consider it as outliers from a country wise point of view, because most of them from dataset seems to be countries within African continent, with limited income composition of resources high life expectancy.

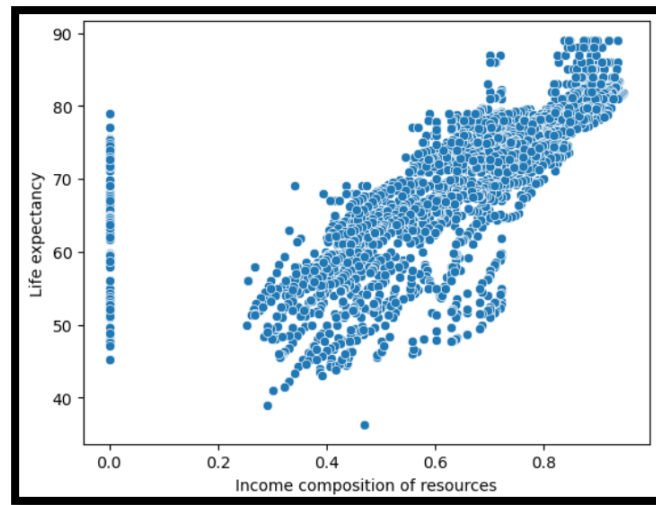


Fig 6. Scatterplot of Income Composition vs Life Expectancy

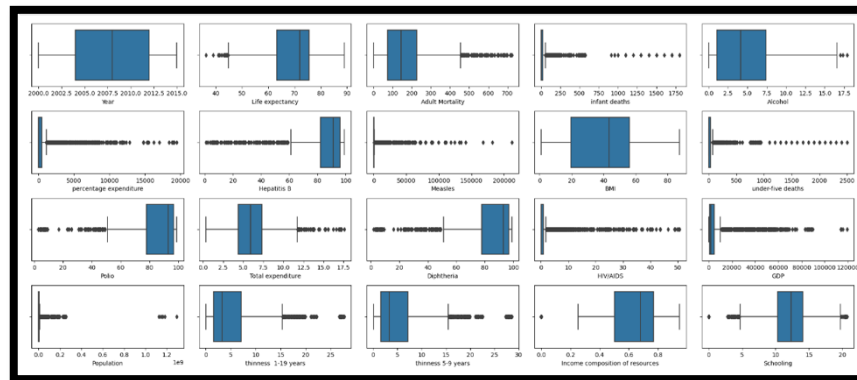


Fig 7. Finding out the outliers in the given dataset using Boxplot

There are many ways to identify and remove outliers: We have implemented Capping datapoints based on IQR (Inter Quartile Range).

```
Q1 = df_copy[cols].quantile(0.25)
Q3 = df_copy[cols].quantile(0.75)
IQR = Q3 - Q1

outliers = ((df_copy[cols] < (Q1 - 1.7 * IQR)) | (df_copy[cols] > (Q3 + 1.7 * IQR))).any(axis=1)

life_expectancy_data = df_copy[~outliers]

life_expectancy_data
```

Fig 8. Applying IQR (Inter Quartile Range) on the dataset to remove any outliers

We are not using this approach as it is resulting in too much loss leaving behind just 1132 observations as we have minimal data. Hence this approach for removing is not feasible.

And since we are building a comprehensive prediction inclusive specific Country, these outliers which appear to us may not be outliers in general.

2.2.3 DEALING WITH CATEGORICAL FEATURES

Here we notice that we have two categorical features namely Country, possessing 193 unique values and Status containing 2 unique values. Hence to implement Multiple Linear Regression and attain coefficients corresponding to these features it is essential to convert to quantitative values.

	Country	Status
1780	Myanmar	Developing
1431	Lao People's Democratic Republic	Developing
74	Antigua and Barbuda	Developing
1757	Morocco	Developing
642	Croatia	Developed

Fig 9. Different categories of data as seen in Status feature: Developed and Developing

Here we used Label Encoder and OHE (One-Hot Encoding) technique so as to preserve the importance of each country in regard to its life expectancy prediction.

```
In [39]: country_dummy
```

Out[39]:

	Albania	Algeria	Angola	Antigua and Barbuda	Argentina	Armenia	Australia	Austria	Azerbaijan	Bahamas	...	United Republic of Tanzania	United States of America	Uruguay	Uzbekistan	Vanuatu
0	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0
16	1	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0
32	0	1	0	0	0	0	0	0	0	0	...	0	0	0	0	0
48	0	0	1	0	0	0	0	0	0	0	...	0	0	0	0	0
64	0	0	0	1	0	0	0	0	0	0	...	0	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2873	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0
2889	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0
2905	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0
2921	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0
2937	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0

2938 rows x 192 columns

Fig 10. One Hot Encoding (OHE) performed on countries and status

In a similar fashion, we extend the application of this process to the 'status' column, wherein the parameter 'drop_first' is set to True. This results in the exclusion of the 'Developed' column composed solely of 0 values. This omission is based on the fact that the presence of 1 values in the 'Developing' column implicitly conveys the information encapsulated by the absence of 1 values in the 'Developed' column. This is seen in the picture below.

```
In [40]: status_dummy
```

```
Out[40]:
```

	Developing
0	1
16	1
32	1
48	1
64	1
...	...
2873	1
2889	1
2905	1
2921	1
2937	1

2938 rows x 1 columns

Fig 11. Developing countries status shown as 1 post OHE

2.3 EXPLORATORY DATA ANALYSIS

The following histogram shows the distribution curve for life expectancy. In this image, for example, there are 64 instances where the life expectancy ranges from 57.5 to 58.4 years.

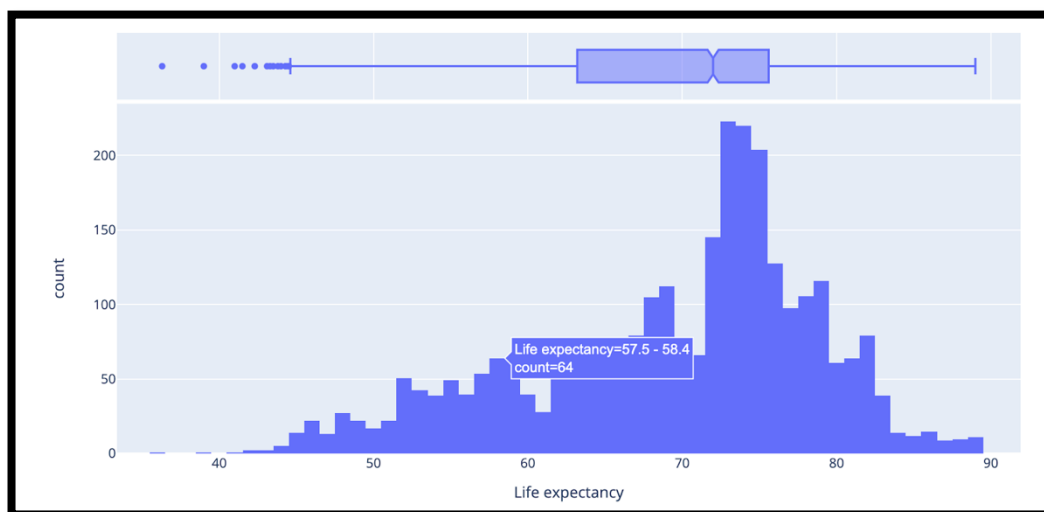


Fig 12. Histogram showing the distribution curve for life expectancy

The next analysis shows a comparison of life expectancy with the status of the countries. The country can either be classified as developed or developing. We know for a fact (subject matter) that developed countries should ideally exhibit high life expectancy. This is seen in the plot as well. The red graph plot (indicating developed countries) has a life expectancy approximately ranging between 68 and 92.

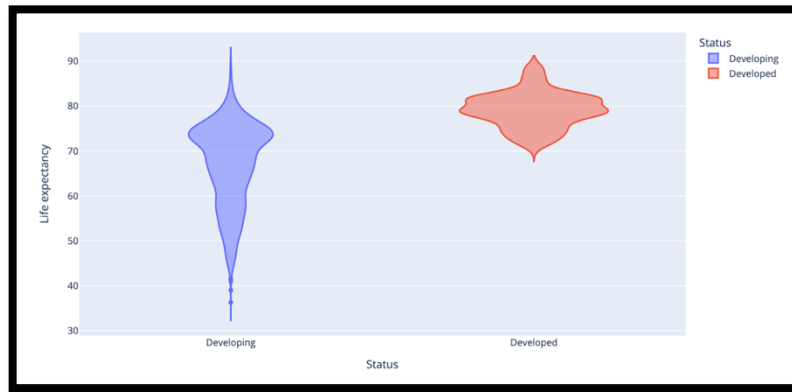


Fig 13. Plot showing the relation between Life Expectancy and Status of the country

Based on our knowledge, we know that if the adult mortality rate is low, the life expectancy should be high. This is seen in the following analysis as well. The red plot depicts developed nations having a higher life expectancy, while adult mortality is on the lower side. The same goes with the comparison of infant deaths as well, which can be seen in the next analysis.

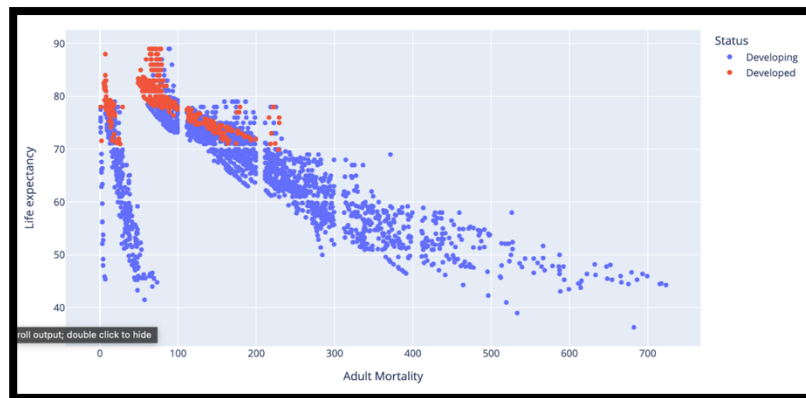


Fig 14. Plot showing the relation between Adult Mortality and Life Expectancy

Life expectancy is seen to be high when infant deaths are fewer. This is seen in the red plot depicting the developed nations. The developed nations have infant deaths almost close to 0.

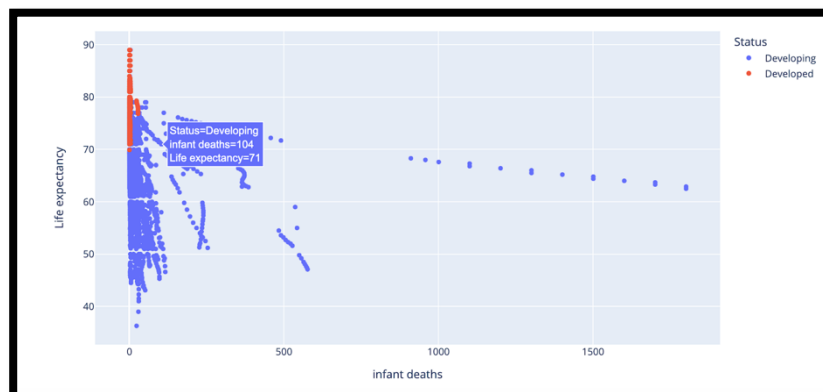


Fig 15. Plot showing the relation between Infant deaths and Life Expectancy

As people go to school, they become more educated, and they are aware of methods to keep their lifestyle healthy. This leads to a higher life expectancy. The more educated people are, the greater their life expectancy. This is seen in the following line-plot below.

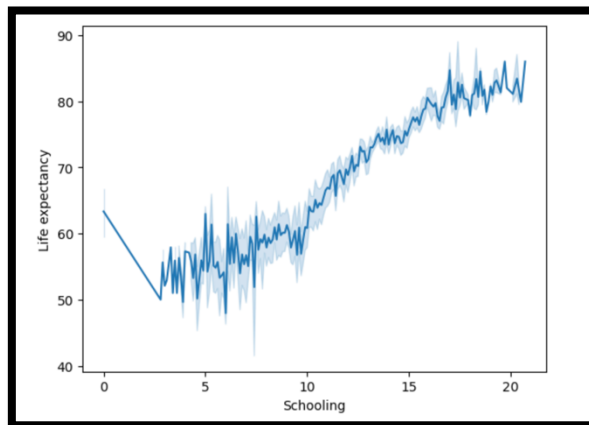


Fig 16. Plot showing the relation between Schooling and Life Expectancy

As expected, the higher Immunization coverage, the higher the life expectancy. The more protected and educated people are about their own health, the higher their life expectancy is. This is seen from the analysis below.

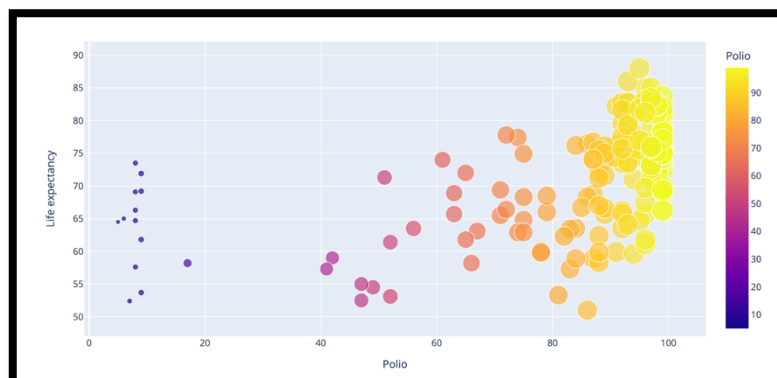


Fig 17. Plot showing the relation between Life Expectancy and Immunization Coverage (Polio)



Fig 18. Plot showing the relation between Life Expectancy and Alcohol Consumption.

It seems that Life Expectancy has none to low relationship with drinking alcohol. From the below scatter plot it is evident that Developed nations consume lot more in quantity compared to Developing nations, but the quality of alcohol consumed too is a factor in determining the life expectancy rather than quantity consumed. The same can be seen in Fig 18.

The total expenditure spent by a nation has very little relation to life expectancy. After analysis of the graphs below, increasing healthcare expenditure does not seem to improve its average lifespan.

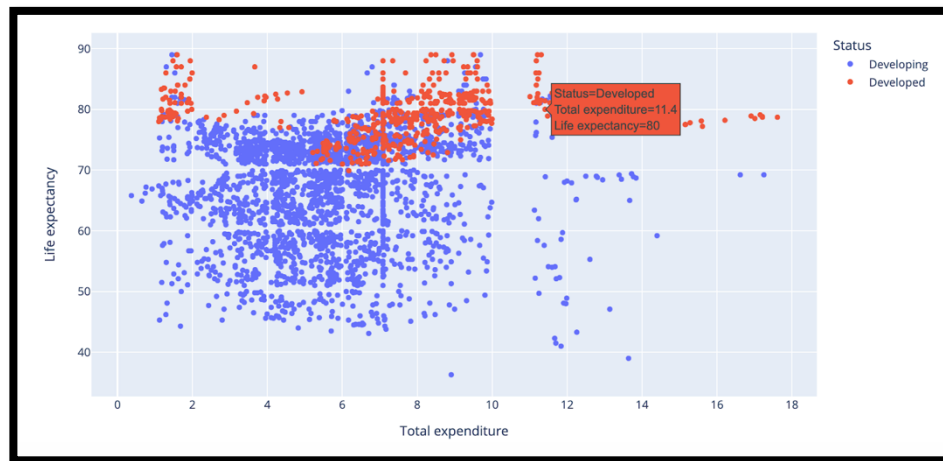


Fig 19. Plot showing the relation between Life Expectancy and Total Expenditure.

The heatmap can be seen below. It takes a correlation between life expectancy and some factors like alcohol, schooling, BMI, and expenditure. It is seen that, apart from schooling, all the other factors have a very low relationship with life expectancy. Only schooling has a relatively high correlation with life expectancy.

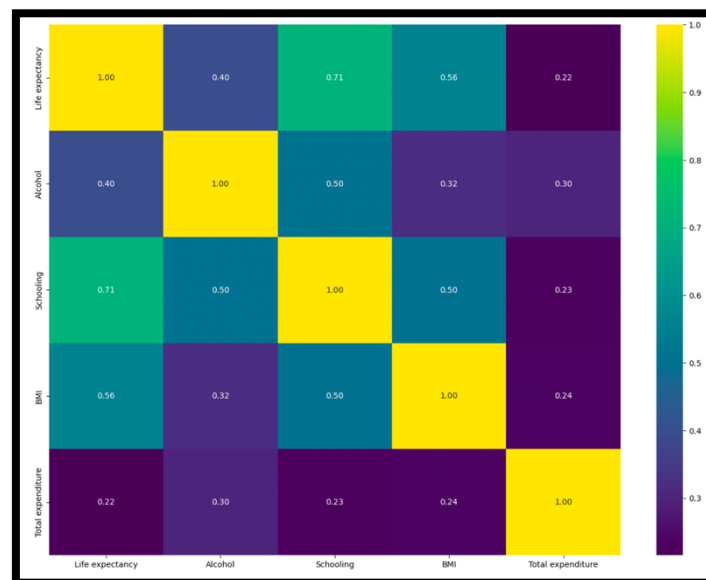


Fig 20. Heatmap showing the relation between Life Expectancy, Schooling, BMI, Alcohol and Expenditure

3. MODEL CONSTRUCTION / REGULARIZATION AND EVALUATION METRICS

3.1 TRAIN – TEST – SPLIT

- ☐ Isolating Model Variables
- ☐ Split data with random state = 0 and 20% test data size
- ☐ Scaling the independent train and test data using Standard Scaler

3.2 MODEL FIT

- ☐ Creation of instance of the Linear Regression class
- ☐ Fit the linear regression model to the training data
- ☐ Predictions on unseen data

```
[ ] Linear_model= LinearRegression()

[ ] Linear_model.fit(X_train,y_train)

LinearRegression
LinearRegression()

[ ] predictions1=Linear_model.predict(X_test)
```

Fig 21. Fitting the Linear Regression model onto the training data

3.3 MEAN ABSOLUTE ERROR

The MAE quantifies the average absolute difference between the actual and predicted values. for each data point in the dataset as shown in Equation 1. MAE provides a straightforward measure of model accuracy.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_{pred,1} - y_{test,1}|$$

Equation 1. Mean Absolute Error

Here we obtain the value as 1.3602491421990355

3.4 MEAN SQUARED ERROR

MSE is the mean of the squared differences between the actual values (y) and the predicted values (y hat) for each data point. It tends to "punish" larger errors more than smaller ones and is widely used in regression evaluation.

Here we obtain the value as 5.868228394132838

3.5 ROOT MEAN SQUARED ERROR

Root Mean Square Error (RMSE) becomes particularly convenient to understand the average magnitude of the errors in the predicted values. It is essentially the square root of the average of the squared differences between predicted values, as reported in Equation 2.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_{pred,i} - y_{test,i})^2}$$

Equation 2. Root Mean Squared Error

Here we obtain the value as 2.4224426503289687

3.6 R² SCORE

A further step for evaluating the prediction capability of the model has been deployed by considering the R² values for each case scenario. This coefficient of determination, defined in Equation 3, is a statistical measure that represents the proportion of the variance in the dependent variable that is predictable from the independent variables in a regression model. Additionally, it is particularly effective since its value range from 0 to 1, where 0 indicates no explanation of variability in the target variable while 1 identifies a perfect variability prediction.

$$R^2 = 1 - \frac{\text{Sum of Squared Residuals}}{\text{Total Sum of Squares}}$$

Equation 3. R²

Here we obtain the value as 0.937132872230424

These metrics allow us to gauge how well the model performs in estimating the target variable. Smaller values of MAE, MSE, and RMSE indicate better model accuracy and a closer fit to the actual data.

```
[ ] from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Calculate Mean Absolute Error (MAE)
MAE = mean_absolute_error(y_test, test_predictions)
MAE

1.3602491421990355

# Calculate Mean Squared Error (MSE)
MSE = mean_squared_error(y_test, test_predictions)
MSE

5.868228394132838

# Calculate Root Mean Squared Error (RMSE)
RMSE = np.sqrt(MSE)
RMSE

2.4224426503289687

# Calculate r2_score
r2_value = r2_score(y_test, test_predictions)
r2_value

0.937132872230424
```

Fig 22. Evaluation metrics performed

3.7 RIDGE REGRESSION

Ridge Regression is a regularization technique that adds a penalty term to the linear regression objective function. This penalty term encourages smaller coefficients for features, helping to prevent overfitting and reducing the impact of collinearity (high feature correlation). RidgeCV is a variant of Ridge Regression that includes built-in cross-validation for alpha (λ) selection. Cross-validation helps us

choose the optimal alpha value by assessing model performance on different alpha values using cross-validation folds.

```
Ridge Regression with Cross-Validated Alpha Selection

[ ] from sklearn.linear_model import RidgeCV

[ ] ridge_cv_model = RidgeCV(alphas=(0.1, 1.0, 10.0), scoring='neg_mean_absolute_error')

[ ] ridge_cv_model.fit(X_train,y_train)

RidgeCV
RidgeCV(scoring='neg_mean_absolute_error')
```

Fig 23. Ridge Regression

3.8 LASSO REGRESSION

Lasso Regression, or L1 regularization, is a linear regression technique that incorporates regularization by adding a penalty term to the linear regression cost function. The penalty term is proportional to the absolute values of the coefficients of the regression variables. LassoCV is being configured to perform Lasso Regression with cross-validated alpha selection.

```
Lasso Regression with Cross-Validated Alpha Selection

[ ] from sklearn.linear_model import LassoCV

lasso_cv_model = LassoCV(eps=0.1, n_alphas=100, cv=5)

[ ] lasso_cv_model.fit(X_train,y_train)

/usr/local/lib/python3.10/dist-packages/sklearn/linear_model/_coordinate_descent.py:1568: DataConversionWarning:
A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples, ), for ex

LassoCV
LassoCV(cv=5, eps=0.1)
```

Fig 24. Lasso Regression

‘alpha’ is the regularization strength, and you can adjust its value based on the degree of regularization you want. Higher values of alpha result in stronger regularization.

3.9 ELASTIC NET

Elastic Net combines the penalties of both Ridge and Lasso Regression

```
Elastic Net with Cross-Validated Alpha Selection combines the penalties of ridge regression and lasso.

[ ] from sklearn.linear_model import ElasticNetCV

elastic_model = ElasticNetCV(l1_ratio=[.1, .5, .7, .9, .95, .99, 1],tol=0.01)

[ ] elastic_model.fit(X_train,y_train)

/usr/local/lib/python3.10/dist-packages/sklearn/linear_model/_coordinate_descent.py:1568:
A column-vector y was passed when a 1d array was expected. Please change the shape of y to

ElasticNetCV
ElasticNetCV(l1_ratio=[0.1, 0.5, 0.7, 0.9, 0.95, 0.99, 1], tol=0.01)
```

	MAE	MSE	RMSE
Ridge Regression	1.3611517836780802	5.85401680448662	2.4195075541288604
Lasso Regression	3.252264936689014	18.489188080055442	4.29990558966769
Elastic Net	1.3930411526783861	6.13343008234664	2.4765762823597095

Fig 25. Elastic Net

Table1. Evaluation Metrics(Regularization)

4. DEPLOYMENT AS WEB APP USING STREAMLIT

4.1. Virtual Environment Anaconda

Install streamlit in Anaconda environment terminal using:

pip install streamlit

Install pickle, for saving the trained model in Anaconda environment terminal using :

pip install pickle-mixin

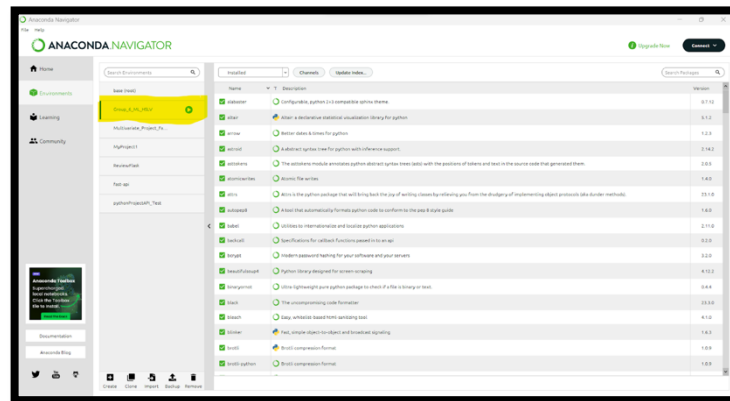


Fig 26. StreamLit loaded into Anaconda Environment

4.4 Saving Trained Model as .sav file

```
[68] ##Saving the simple trained model for Web UI
import pickle
filename = 'general_life_expectancy_ui.sav'
pickle.dump(Linear_model_web_ui, open(filename, 'wb'))
```

Fig 27. Saving the trained model as a .sav file

4.5 Using Spyder IDE to build User Interface (streamlit library in python)

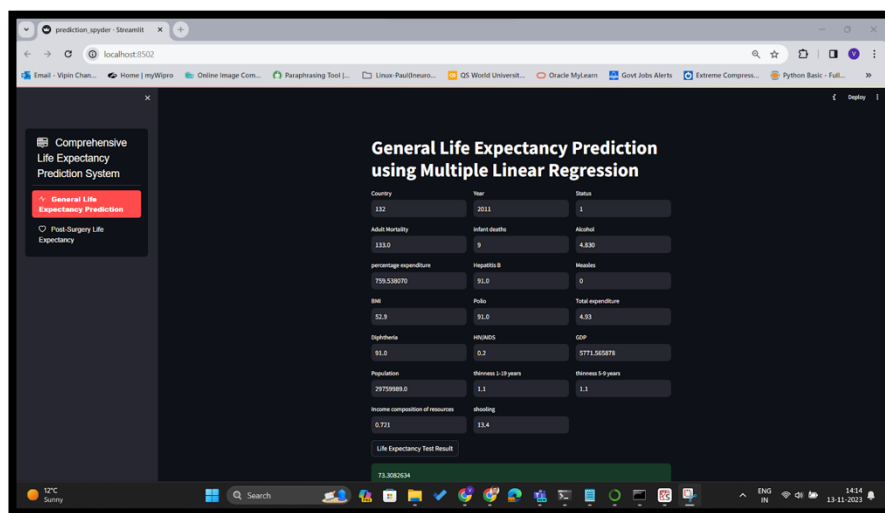


Fig 28. User Interface of the Web App deployed using StreamLit